

Summary

Exquisite learner with a strong academic foundation in computer engineering and software development. Passionate about hands-on projects, problem-solving, and collaborative work. Actively seeking opportunities to apply technical knowledge while strengthening real-world development and analytical skills.

Education

Padre Conceicao College of Engineering, Verna, Goa

Bachelor of Engineering in Computer Engineering | 2023 – 2027

Current CGPA (Sem I – Sem IV): 9.2

Shri Damodar Higher Secondary School of Science, Margao, Goa

HSSC | 2023

Percentage: 70%

Mahila and Nutan English High School, Margao, Goa

SSC | 2021

Percentage: 91.83%

Skills

Programming Languages

- C
- C++ (Data Structures)
- Java
- Python
- JavaScript

Web Technologies

- HTML5
- CSS
- JavaScript
- Node.js
- PHP

Databases & Backend

- MySQL
- MongoDB
- Supabase

Certifications

Courses

- Data Structures and Algorithms in C – Udemy (2024–2025)
- Software Testing – NPTEL, IIT Bangalore (2025, 3 Credits)
- Understanding Incubation and Entrepreneurship – NPTEL, IIT Bombay (2025, 3 Credits)

Internships

- Deloitte Virtual Internship Program Deloitte | May 2025
- Introduction to Generative AI Virtual Internship Google Cloud | July 2025
- GenAI-Powered Data Analytics Virtual Internship Tata | September 2025

Projects

Completed Projects

Urban Gardening – Full Stack Application

July 2025 – August 2025

- Designed a full-stack platform to help users learn and implement home gardening practices.
- Implemented plant data storage and care-related insights for users.
- Developed a responsive frontend using HTML, CSS, and JavaScript, with Supabase-based user authentication.
- Integrated backend services using Node.js with MySQL database connectivity.
- Promotes sustainable living through an intuitive digital solution.

Technologies Used: HTML5, CSS3, JavaScript, Node.js, MySQL, Supabase (Authentication)

<https://github.com/URBAN-GARDENING>

TuneVault – Full Stack Application

August 2025 – October 2025

- Built a full-stack web portal for registering and managing song metadata.
- Designed a digital trademark-style registration system for artists, songs, and albums.
- Implemented role-based access control to ensure only owners can update or delete registered songs.
- Developed features for artist registration, song registration, and album creation.
- Focused on clean architecture and scalable backend design.

Technologies Used: HTML5, CSS3, JavaScript, Node.js, MySQL, Supabase

<https://github.com/SONG-REGISTRATION>

Online Recipe Book – C++

July 2024 – September 2024

- Designed and implemented a console-based Recipe Management System using C++ and OOP principles.
- Applied inheritance and runtime polymorphism to support multiple recipe categories with reusable logic.
- Integrated file handling (fstream) to persist recipes, user credentials, and categories.
- Developed a user authentication system supporting account creation and login validation.
- Built a menu-driven interface for adding and viewing recipes efficiently.

Technologies Used: C++, Object-Oriented Programming (OOP), File Handling (fstream)

<https://github.com/Online-Recipe-Book>

Student Management System – Java

October 2025 – November 2025

- Developed a Java-based Student Management System using object-oriented programming principles.
- Implemented full CRUD functionality for managing student records.
- Structured the application using Java packages and classes to separate concerns.
- Integrated a MySQL database with JDBC for persistent data storage.
- Designed search and listing features for efficient student record retrieval.

Technologies Used: Java, JDBC, MySQL, Object-Oriented Programming (OOP)

<https://github.com/JAVA-STUDENT-MANAGEMENT>

Ongoing Projects

Automotive Black Box with Accident Detection & Prediction using ML – BITS PILANI GOA

- Designing a user-owned automotive black box system to record, analyze, and preserve pre-crash, crash, and post-crash vehicle data.
- Implementing real-time vehicle data acquisition from the OBD-II interface using CAN bus (MCP2515 + transceiver) integrated with an ESP32 controller.
- Integrating multi-sensor data sources including IMU (accelerometer/gyroscope) and GPS for motion analysis, impact detection, and time-synchronized event reconstruction.
- Developing ESP32 firmware in C/C++ for real-time acquisition of CAN bus, IMU, and GPS data using UART, SPI, and I²C interfaces.
- Designing edge-based accident risk prediction using sensor fusion, time-series feature extraction, and lightweight models suitable for embedded (ESP32) deployment with a privacy-first approach.

TRAVELLOG – Full Stack Application

- Developed a full-stack web application that enables users to create, store, and share digital travel journals including images, blog content, travel tips, locations, and map references.
- Implemented secure user authentication and authorization using Supabase Auth, supporting user registration, login, and session management.
- Designed and integrated MongoDB as the primary database to manage travel logs, user-generated content, ratings, and metadata efficiently.
- Built a real-time community chat feature using Socket.IO, enabling users to create chat rooms and participate in live discussions with other travelers.
- Focused on clean architecture and scalable backend design.

Technologies Used: HTML5, CSS3, JavaScript, Node.js, MongoDB, Socket.IO

SMART WATER MONITORING – IDEAS PROJECT

- Designing an IoT-based water quality monitoring system to track critical parameters such as pH, turbidity, and Total Dissolved Solids (TDS) in real time for rural water sources.
- Building a centralized Web application to display live water quality data with intuitive safe/unsafe indicators for easy understanding by non-technical users.
- Exploring the use of machine learning models to predict potential contamination events based on seasonal changes, rainfall, and environmental factors.
- Developing an affordable and scalable sensor architecture
- Collaborating in a team-based engineering environment, contributing to system design, implementation planning, and testing strategies.

Strengths

- Strong problem-solving and analytical thinking skills
- Quick learner with a solid academic foundation
- Ability to work effectively both independently and in teams
- Strong understanding of end-to-end web application development, from frontend UI design to backend logic and database integration.

Declaration

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

Yours sincerely,

Swayam Prabhu